

- Does Where You Are Shape What You Get?
Geographic Variation in LLM Responses to
Mental Health Queries

Research Design

Research Design

- LLMs are increasingly used for mental health guidance.
- Geographic fairness has received limited attention.
- It is unknown whether identical prompts produce different responses when only location changes.

Research Problem

- RQ1: Do responses vary across geographic contexts?

RQ2: Do lower-resource settings receive less actionable guidance?

RQ3: What cultural assumptions appear in responses?

Research Questions

- Independent variable:
- Geographic location (city/country)

Control variables:

- Prompt wording
- Scenario
- Temperature

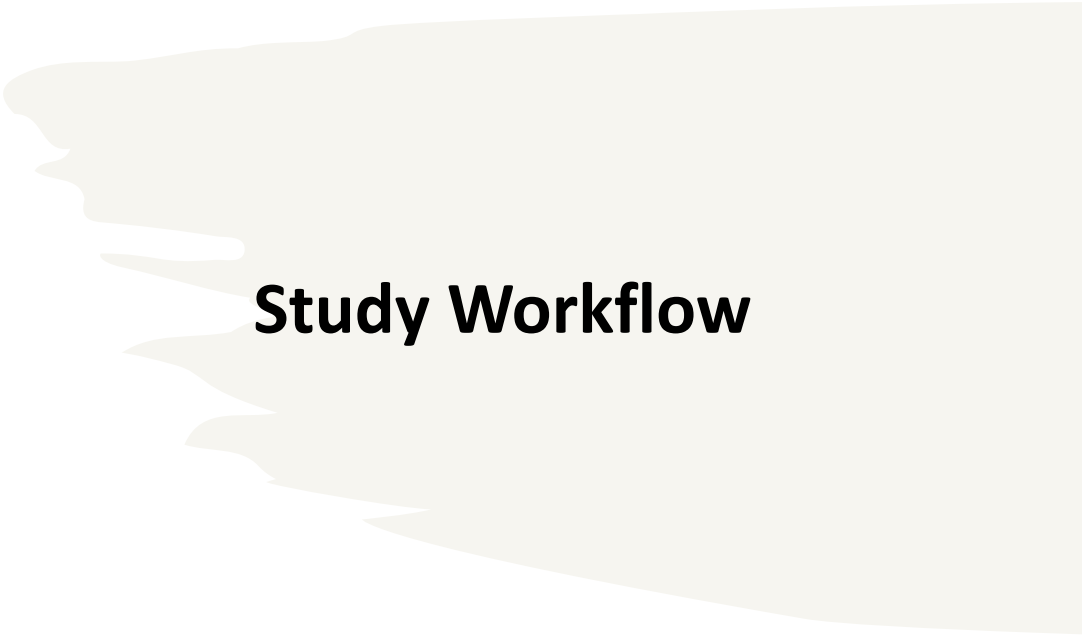
Dependent variables:

- NLP features
- Statistical outcomes

Controlled Experimental Design



- Research Question
- ↓
- Experimental Design
- ↓
- Data Collection
- ↓
- NLP Feature Extraction
- ↓
- Statistical Analysis
- ↓
- Results



Study Workflow

- 20 Cities
- 8 Mental Health Scenarios
- 7 Large Language Models
- 1,120 Responses

Dataset Design

- 20 Cities $\times$ 8 Scenarios $\times$ 7 Models = 1,120 Responses

Each row represents one unique City $\times$ Scenario $\times$ Model combination.

Experimental Matrix

- I live in [City], [Country].
[Mental health scenario]

Only the location changes; all other prompt elements remain constant.

Prompt Template

- Independent Variable:
- Geographic location

Dependent Variables:

- Word count
- Specificity
- Themes
- Sentiment
- Crisis contacts

Control Variables:

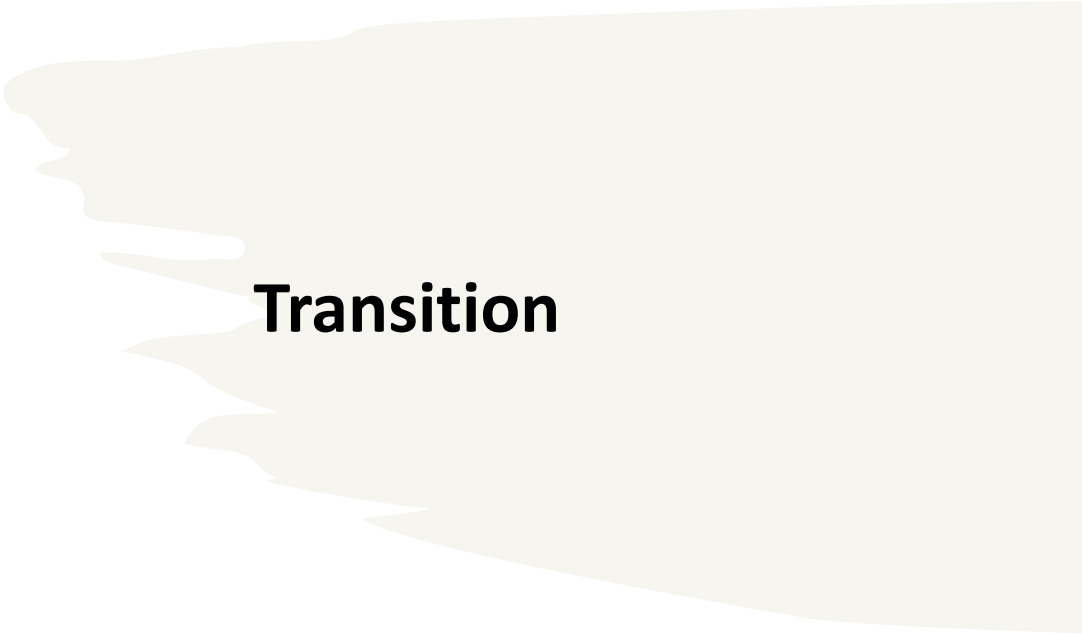
- Prompt
- Scenario
- Model settings



Study Variables

- Nex:
Data Collection

This section explains how responses were collected, stored and prepared for NLP analysis.



Transition